

Pi-CloudDOOM — InterviewPrep TN

Microservice platform for interview preparation, training, mentorship, and community features. Polyglot stack: Spring Boot backends + FastAPI AI services + Angular frontend, fronted by Keycloak.

Tech Stack

- **Backend:** Spring Boot 4.x (Java 21), 7 microservices
- **AI:** FastAPI (`ai-training-path`), Whisper (STT), Kokoro (TTS), Ollama (llama3.2:3b)
- **Frontend:** Angular 21
- **Auth:** Keycloak 24
- **Data:** PostgreSQL 16, Redis 7, Kafka 7.5, MinIO (S3-compatible)
- **Migrations:** Flyway 11
- **Mail (dev):** MailHog (Mailtrap-compatible)

Services

Service	Port	Purpose
user-service	8081	Users, profiles, RBAC
interview-service	8082	Mock interview sessions
training-service	8083	Training paths, gamification, XP
mentorship-service	8084	Mentorship matching
quiz-service	8085	Quizzes, oral attempts
community-service	8086	Community feed
resource-service	8087	Learning resources
ai-training-path	8001	ML/LLM-driven training path generation
frontend (Angular)	4200	Web UI
Keycloak	8080	OIDC + realm <code>myapp-realm</code>

Quick Start (local, Docker)

1. Prerequisites

- Docker 24.0 + Docker Compose 2.20
- ~6 GB free RAM (the full stack runs ~4–5 GB)
- Java 21 + Maven 3.x **only if** you want to run a service outside Docker

2. Configure environment

```
git clone <this-repo>
cd PULL
cp infra/.env.example infra/.env
```

Open `infra/.env` and set the **required** API keys (free tiers, ~30s sign-up each):

Variable	Required?	What it powers	Get one at
GROQ_API_KEY	Yes	interview-service and community-service refuse to start without it	https://console.groq.com/keys
GOOGLE_AI_API_KEY	Yes	training Coach chatbot in the frontend (button stays disabled if blank, no crash)	https://aistudio.google.com/app/apikey
MAIL_USERNAME	Optional	Real SMTP for <code>/api/emails/test</code> (else MailHog catches everything)	Mailtrap sandbox
MAIL_PASSWORD			

Important — Postgres password is baked into the volume on first run. The value of `POSTGRES_PASSWORD` is stored in the `infra_pgdata` volume at first `docker compose up`. Changing `.env` afterward will **not** rotate the actual password — Keycloak and the Spring services will fail with **FATAL: password authentication failed for user "postgres"**. Pick the password you want **before** the first up; if you need to reset, run `docker compose down -v` (wipes all DB data).

3. Start the stack

```
cd infra
docker compose up -d
docker compose ps
```

First boot takes ~90 s (Keycloak imports the realm). Wait for all services to show `healthy` or `Up`.

4. Verify

Endpoint	What it shows
<code>http://localhost:4200</code>	Angular frontend
<code>http://localhost:8080/realms/myapp-realm</code>	Keycloak realm metadata
<code>http://localhost:8080/admin/</code>	Keycloak admin console
<code>http://localhost:8081/actuator/health</code>	user-service health
<code>http://localhost:5050</code>	pgAdmin (admin@example.com / admin123)
<code>http://localhost:8025</code>	MailHog inbox

5. Seed test users (one-time)

The realm import doesn't ship with shareable test users (only the team's personal accounts). Run the seeding script once after the stack is up:

```
infra/scripts/seed-test-users.sh
```

This creates three users via the Keycloak admin API: - admin@test.com / admin123 — ROLE_ADMIN - user@test.com / user123 — ROLE_USER - mentor@test.com / mentor123 — ROLE_MENTOR (mentorship-service endpoints)

The script is idempotent — re-running it skips users that already exist.

6. Test login

Open <http://localhost:4200>, click **Login**, use one of the seeded accounts above, or click **Register** to create your own.

API Reference (selected — user-service)

Method	Endpoint	Auth
POST	/api/users/register	Bearer token
GET	/api/users/me	Bearer token
GET	/api/users	ADMIN
GET	/api/users/{id}	ADMIN/owner
PATCH	/api/users/{id}/role	ADMIN

Postman collections at the repo root: PI-user.postman_collection.json, PI-training-personalization.postman_collection.json, member4-training-crud.postman_collection.json.

Repository layout

```
.
infra/                                # docker-compose, Keycloak realm, init SQL, .env.example
  docker-compose.yml
  .env.example                        # ← copy to .env before first up
  keycloak/                          # realm-export.json, themes
  db/                                # init-multiple-databases.sh
  scripts/                           # m4_smoke.sh
frontend/                            # Angular 21 app
user-service/                        # Spring Boot
interview-service/
training-service/
mentorship-service/
```

```

quiz-service/
community-service/
resource-service/
ai-training-path/      # FastAPI
kokoro/                # TTS
vercel.json            # frontend deploy config

```

Deployment

Production deployment is hybrid:

Component	Platform
Spring microservices, Keycloak, Postgres, Redis, Kafka, MinIO	OpenStack Kubernetes (picloudoom namespace)
AI services (ai-training-path, Whisper, Kokoro, Ollama)	Azure Container Apps (France Central)
Angular frontend	Vercel
Container images	DockerHub (docker.io/azizbna/pi-clouddoom-*)

CI/CD is fully automated via GitHub Actions — see [docs/cicd/README.md](#) for the pipeline diagram, workflow index, secret matrix, and runbook. OpenStack cluster IaC lives at [infra/openstack/heat-cluster.yaml](#).

Branch structure

- **main** — released code
- **integration/final** — current integration branch (you are here)
- **deployment/hybrid-openstack-azure-vercel** — deployment manifests (created in deployment phase)

Troubleshooting

Symptom	Likely cause	Fix
Browser: ERR_CONNECTION_RESET on Keycloak (8080)	Keycloak crash-looping on Postgres auth	See “Postgres password baked in volume” warning above. Run <code>docker compose logs userservice-keycloak</code> .
Spring services restart loop	Same as above — wrong DB password	Same fix.

Symptom	Likely cause	Fix
Port 8080/4200/5432 already in use	Another process bound	Edit the corresponding <code>*_PORT</code> in <code>infra/.env</code> and re-up.
Topics Kafka non créés	<code>kafka-init</code> ran before Kafka was healthy	<code>docker compose restart kafka-init</code> and check its logs.
Frontend builds but shows blank page	Wrong <code>KEYCLOAK_ISSUER_URI</code> or stale browser cache	Hard reload (Ctrl+Shift+R), check <code>frontend/src/environments/environment.ts</code> .

For more, see `infra/README.md`.